TRANSITIVITY OF THE UNIVERSAL THICKENING AND CONORMAL SEQUENCE FOR FORMALLY UNRAMIFIED MORPHISMS OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3425-3445 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 3447-3469, the complete author proof (21 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Universal first order thickenings of formally unramified morphisms of algebraic spaces and conormal sheaves; the construction is retained.

Source reference. Original lemma 04CQ: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 557–570. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06AE: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 591–603. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04F7: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3218–3241. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $Z \to Y \to X$ be formally unramified morphisms of algebraic spaces over S.*

(1) *If $Z \subset Z'$ is the universal first order thickening of Z over X and $Y \subset Y'$ is the universal first order thickening of Y over X, then there is a morphism $Z' \to Y'$ and $Y \times_{Y'} Z'$ is the universal first order thickening of Z over Y.*

(2) *There is a canonical exact sequence*

$$i^*\mathcal{C}_{Y/X} \to \mathcal{C}_{Z/X} \to \mathcal{C}_{Z/Y} \to 0$$

where the maps come from Lemma 04F7 and $i : Z \to Y$ is the first morphism.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. The map $h : Z' \to Y'$ in (1) comes from Lemma 04F7. The assertion that $Y \times_{Y'} Z'$ is the universal first order thickening of Z over Y is clear from the universal properties of Z' and Y'. By Lemma 06AE we have an exact sequence

$$(i')^*\mathcal{C}_{Y\times_{Y'}Z'/Z'} \to \mathcal{C}_{Z/Z'} \to \mathcal{C}_{Z/Y\times_{Y'}Z'} \to 0$$

where $i' : Z \to Y \times_{Y'} Z'$ is the given morphism. By Lemma 04CQ there exists a surjection $h^*\mathcal{C}_{Y/Y'} \to \mathcal{C}_{Y\times_{Y'}Z'/Z'}$. Combined with the equalities $\mathcal{C}_{Y/Y'} = \mathcal{C}_{Y/X}$, $\mathcal{C}_{Z/Z'} = \mathcal{C}_{Z/X}$, and $\mathcal{C}_{Z/Y\times_{Y'}Z'} = \mathcal{C}_{Z/Y}$ this proves the lemma. □